

EE143: Microfabrication Technology

Spring 2026

Lab Section: Wednesday 9-12 PM

Group 20

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GSI: Shiwoo Lee

Professor: Ali Javey

Contributions:

Huan: Summary [I], inputted layouts, did measurements graphs and tables, and did all the calculations for [II, III, IV, V], inputted some layouts, some measurement graphs and some tables, helped with discussion [VI]. Provided math work for appendix.

Prathamesh: Created Graphs for MOSFETS [VI] and extracted the V_T using the graphical method for all the devices as well (in appendix), and calculated I_{on} to I_{off} ratios. Formatted the final document.

Yarizmar: Provided the overview and discussion of all portions and then did the entire inverter section [I, II, II, IV, V, VI, VII] and calculated yield.

I. Summary

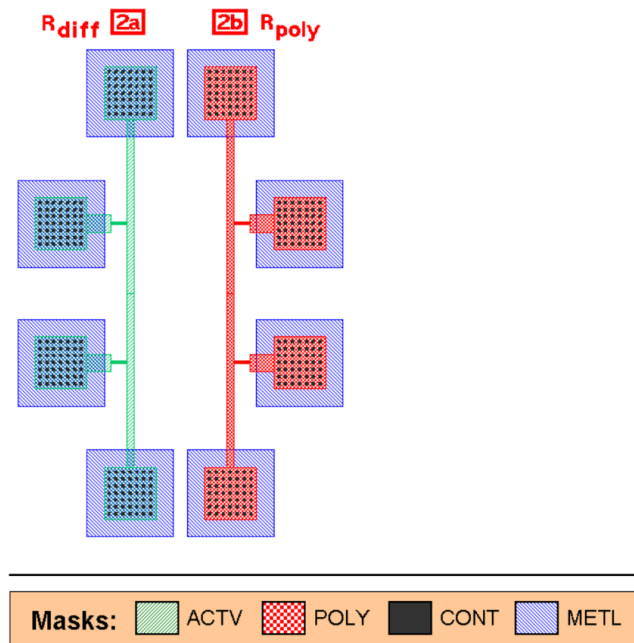
The overall goal of the characterization portion of our process flow is to test and measure the electrical behavior of fabricated devices on our wafer, including resistors, capacitors, diodes, and MOSFETs. During this process, we focused on evaluating how closely their real performance matches theoretical expectations and design parameters. We were able to successfully extract all of the data needed, like I-V curves for resistors, C-V curves for capacitors, and I_d - V_d / I_d - V_g curves for MOSFETs. Overall, the measurements appeared to be accurate for simpler structures like resistors and some MOSFETs, where trends such as linear I-V behavior or expected saturation regions were clearly observed. However, some devices didn't work as expected: our resistors' contacts weren't etched well, so we decided to probe the middle contacts as well as find a good MOSFET to probe since it was giving us some bad curves. Through this process, we learned how sensitive device performance is to fabrication steps, how to interpret real (non-ideal) data, and how to connect measured characteristics back to physical device properties.

II. 4-Point Resistors (2a, 2b)

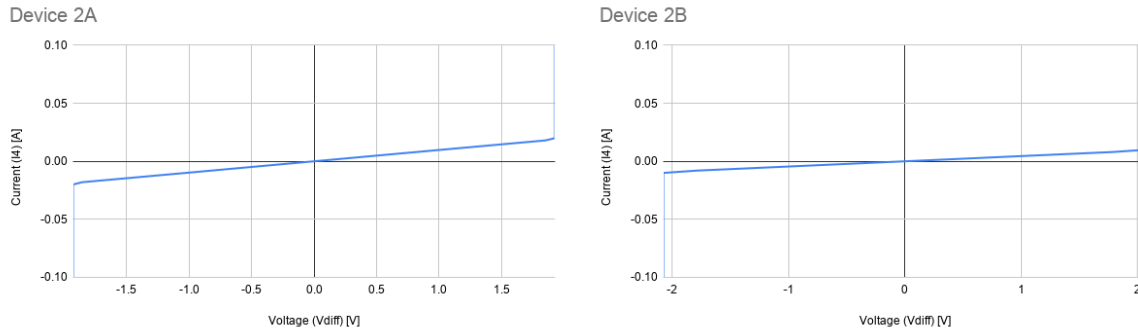
Overview:

The 4-Point Resistors are test structures that are formed in different process layers during fabrication, and can be used to measure the sheet resistances of the ACTV and POLY layers. R_{diff} is formed during the diffusion/active layer and R_{poly} is formed during the polysilicon layer. The active layer sheet resistance is dependent on doping areas, junction depth, and carrier motion, and comparing R_{diff} with the theoretical resistance for these steps allows us to determine the success of our process. The same is true for R_{poly} where the POLY layer's resistance is dependent on doping and thin film thickness. Both are measured using the 4 probes on the probing station, where current is forced through the two outer contacts and the probes attached to the inner contacts measure the voltage drop. This process eliminates the effect of the resistance of the contacts and probes. We generally expect that R_{diff} will be larger than R_{poly} .

Layout:



Measurements:



Parameter	Value
Measured resistance of 2a (Ω)	102.48 Ω
Measured resistance of 2b (Ω)	222.33 Ω

Calculations:

Sheet resistance of phosphorus-doped substrate: 3 Ω /square

- sheet resistance of polysilicon (also phosphorus-doped): 3 Ω /square
- polysilicon thickness: 450nm
- junction depth in phosphorus-doped regions: 1.0 μm
- ACTV layer overetch: 0.5 μm (this makes ACTV features larger)
- POLY layer overetch: 0.5 μm (this makes POLY features smaller)

Parameter	Value
Length of resistor 2a, including overetch (μm)	201 μm
Width of resistor 2a, including overetch (μm)	11 μm
Height of resistor 2a (μm)	1 μm
Theoretical resistance of 2a (Ω)	54.8 Ω

Parameter	Value
Length of resistor 2b, including overetch (μm)	199 μm
Width of resistor 2b, including overetch (μm)	9 μm
Height of resistor 2b (μm)	0.45 μm
Theoretical resistance of 2b (Ω)	66.3 Ω

Discussion:

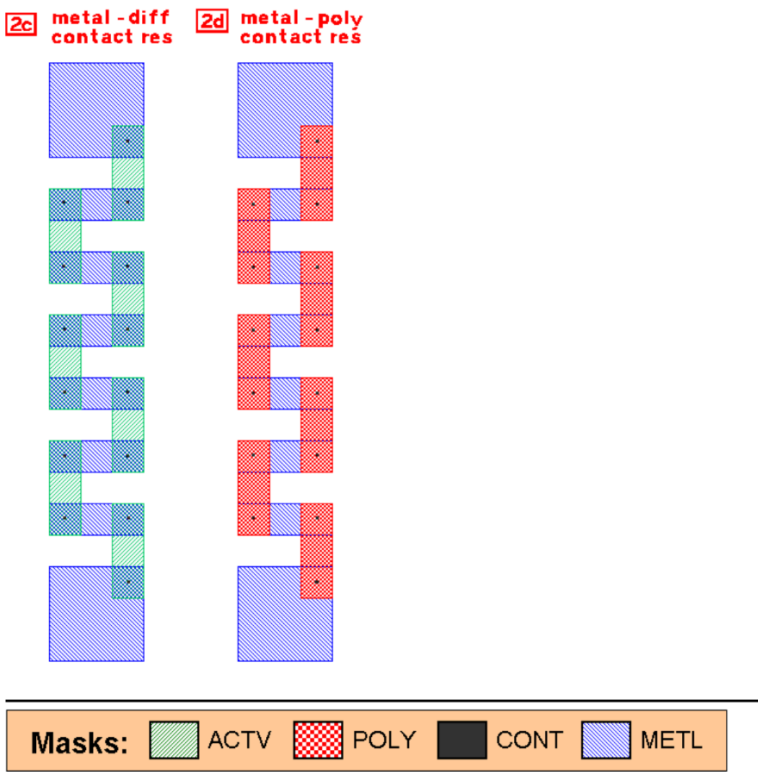
The measured resistances obtained for both 2a and 2b were much higher than the theoretical values calculated. It did, however, follow the logic of R_{poly} being a higher resistance than R_{diff} . The most probable reason for the discrepancies is from poor probe measurements from outdated equipment and contamination during the doping and etching process, though it could also be the result of the fact that the fabricated structures might have differing dopant concentration, junction depth, and polysilicon thickness. The difference between the measured and theoretical resistances were 11.5 Ω , as the theoretical resistance was expected to be higher.

III. Contact Chain Resistors (2c and 2d)

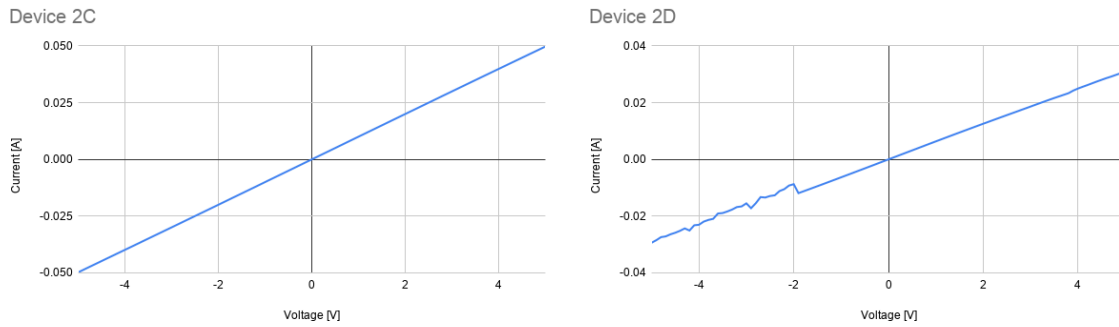
Overview:

The purpose of contact chain resistors is to ensure that there is uniformity and structural integrity within the conductive regions of the wafer. If any portion of the conductive regions are dysfunctional then there is an open circuit. The chain is measured by touching the beginning of the chain at the top and then one of the inner, middle chains. The difference between the diff and poly contact chain resistors is the type of conductive metal utilized.

Layout:



Measurements:



Parameter	Value
Measured resistance of 2c (Ω)	99.96 Ω
Measured resistance of 2d (Ω)	157.64 Ω

Calculations:

We can use these devices to measure the resistance of a metal-polysilicon or metal-substrate contact by subtracting the known resistances of everything else (i.e., metal, polysilicon, and doped substrate areas). Complete the following table using dimensions from the layout and assuming the following:

- sheet resistance of phosphorus-doped substrate: 3 Ω /square
- sheet resistance of polysilicon (also phosphorus-doped): 3 Ω /square
- sheet resistance of aluminum: 0.3 Ω /square

Parameter	Value
Resistance of a metal chain link (Ω)	0.9 Ω
Resistance of an ACTV area chain link (Ω)	9 Ω
Resistance of a polysilicon chain link (Ω)	9 Ω
Expected resistance of 2c without contacts (Ω)	69.3 Ω
Expected resistance of 2d without contacts (Ω)	69.3 Ω

Resistance of a single 2c contact (Ω)	2.19 Ω
Resistance of a single 2d contact (Ω)	6.31 Ω

Discussion:

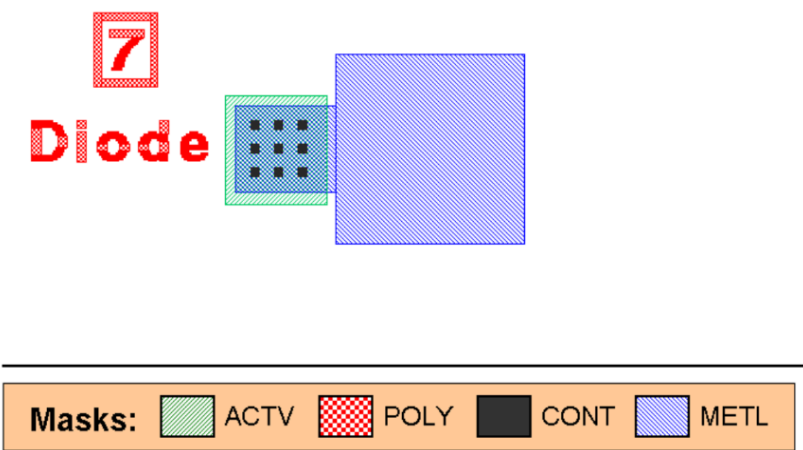
The measured resistance was about double than the expected resistance. The 2d device did have a general trend, but the values attained were a lot lower than expected. There is a slight discrepancy in the beginning of the graph which likely contributed to the skew in the data. This likely resulted from poor probe placement or contamination on the resistive portions which would increase the overall resistance. The probes on the device had bends and made it difficult to place exactly where we intended to. The environment in which we created the wafer's characteristics was less than optimal clean conditions.

IV. Diode (7)

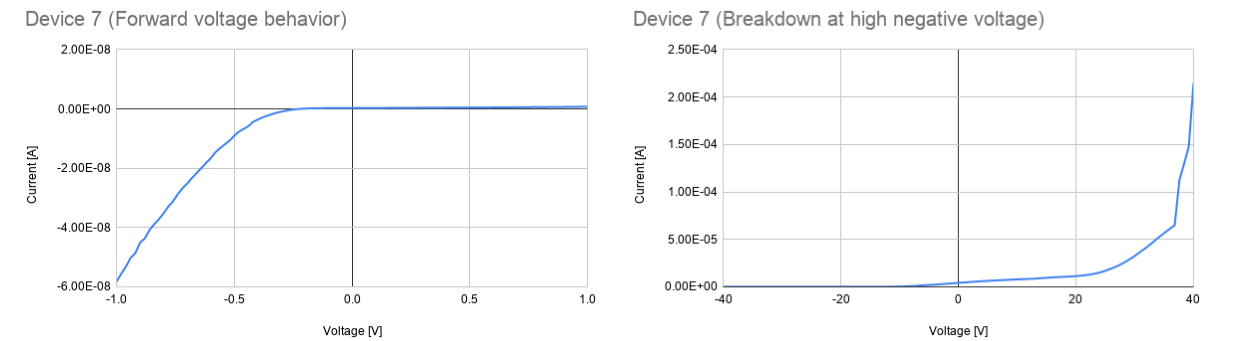
Overview:

The diode is utilized to send current into a singular direction and turn alternating current into direct current which is especially useful for maintaining a constant current throughout a device. This protects sensitive components in the wafer from getting too much voltages and short circuiting. This measurement is taken by placing a probe onto the diode whilst a secondary probe is a constant at the stageconnector placed directly onto the chuck.

Layout:



Measurements:



Parameter	Value
Forward threshold voltage (V)	~ 0.3 V
Breakdown voltage (V)	~ 37.6 V

Calculations:

It is possible to approximate the diode threshold voltage, sometimes also called the built-in potential, given only the amounts of N and P doping on the sides of its p-n junction (see the lecture slides on devices or, e.g., chapters 1 and 4 of the textbook "Modern Semiconductor Devices for Integrated Circuits" by Chenming Hu at <https://www.chu.berkeley.edu/modern-semiconductor-devices-for-integrated-circuits-chenming-c-alvin-hu-2010/>). Do this assuming the following values from fabrication, and remembering you can use charts to convert between silicon bulk resistivity and dopant density:

- bulk resistivity of phosphorus-doped N side of junction: $1\text{e-}3\ \Omega\text{-cm}$
- bulk resistivity of P side of junction: $1\ \Omega\text{-cm}$

Parameter	Value
p-type dopant density N_A (# dopant atoms / cm^3)	$1.3 \times 10^{16}\ \text{cm}^{-3}$
n-type dopant density N_D (# dopant atoms / cm^3)	$4.6 \times 10^{18}\ \text{cm}^{-3}$
Threshold voltage / built-in potential (V)	0.88 V

Discussion:

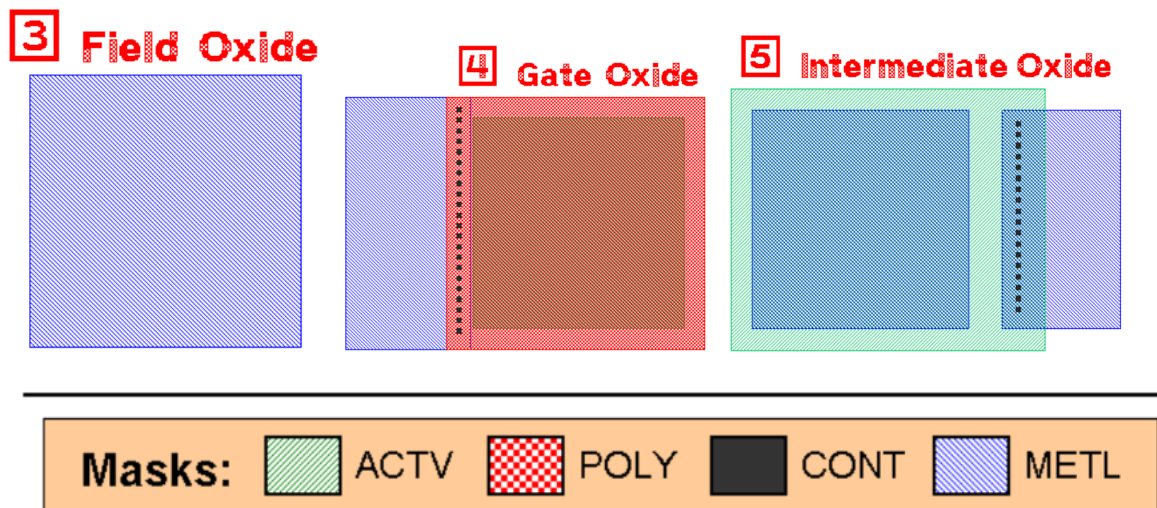
The forward and threshold voltage are found to be not very similar because the value is not close in range. Whilst the forward threshold voltage is 0.6 V the threshold voltage is 0.88 V theoretical making for a 0.28 V difference. This leads to a margin of error of approximately 31% which may have resulted due to higher resistances found in the components of the wafer which lowered the voltage. Possible causes for the error is poor probe placement and bent probes in addition to contact resistance errors which involve contamination and bad etching/processing.

V. Capacitors (3, 4, 5)

Overview:

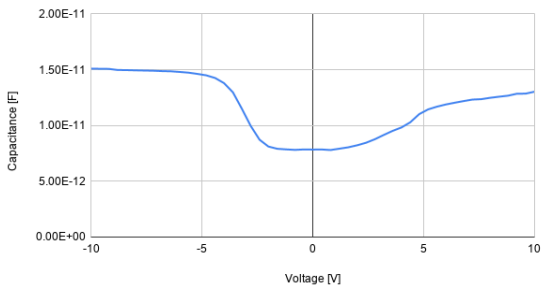
The purpose of a capacitor is to store electric charge, which is created from metal, oxide, and semiconductor in the microfabrication process. The capacitor is important for controlling the current, and it was found in each by placing a probe into the designated oxide, and then the other probe was placed upon the chuck with the field oxide getting measured twice (one with a light source and one without). The gate oxide is utilized between the gate and the semiconductor as a thin insulating material. Meanwhile, the field oxide is grown on the silicon substrate to aid in ensuring active devices are isolated from one another and avoid short-circuits and meshing. This is achieved by being a generally thicker layer than that found in the gate oxide. The intermediate oxide is usually considered a layer that performs as a back-up for the gate oxide, where it prevents leakage, ensures all properties act accordingly, and avoids taking current from other sources.

Layout:

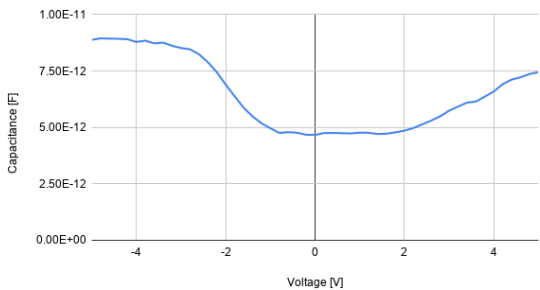


Measurements:

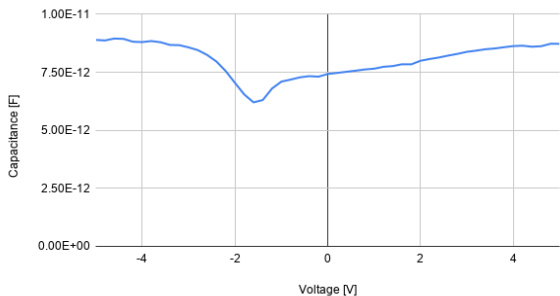
Device 3



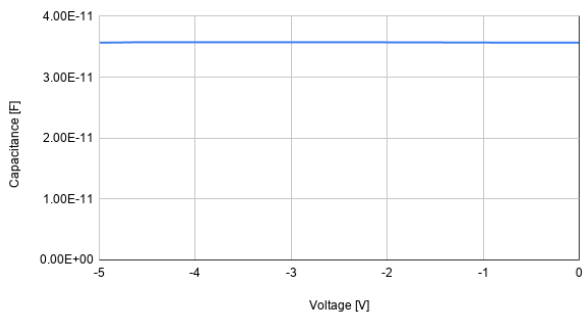
Device 4 (No Light)



Device 4 (With Light)



Device 5



Parameter	Value
Field oxide capacitor capacitance (in accumulation) (pF)	1.5 pF
Field oxide capacitor flatband voltage (V)	~ -4.5 V
Field oxide capacitor threshold voltage (V)	~ -1 V

Parameter	Value
Gate oxide capacitor capacitance (in accumulation) (pF)	8.94 pF
Gate oxide capacitor flatband voltage (V)	~ -4 V
Gate oxide capacitor threshold voltage (V)	~ -1 V

Parameter	Value
Intermediate oxide capacitor capacitance (pF)	3.57 pF

Calculations:

We can compute theoretical values for capacitance with the parallel plate capacitor approximation. Find the requested values using the device layout and assuming the oxide parameters were as follows (note these field and intermediate oxides are thinner than originally fabricated because they were partially etched by HF several times):

- field oxide thickness: 200nm
- gate oxide thickness: 50nm
- intermediate oxide thickness: 100nm
- SiO₂ relative permittivity: 3.9

Parameter	Value
Field oxide capacitor area (μm^2)	40000 μm^2
Gate oxide capacitor area (μm^2)	40000 μm^2
Intermediate oxide capacitor area (μm^2)	40000 μm^2
Theoretical field oxide capacitor capacitance (pF)	6.903 pF
Theoretical gate oxide capacitor capacitance (pF)	27.61 pF
Theoretical intermediate oxide capacitor capacitance (pF)	13.81 pF

Discussion:

The capacitors did somewhat exhibit the exact graph behavior found in most capacitors. It had the general dip and rise in all the graphs. The measured and theoretical values, however, varied tremendously across the board. While it was expected to get 27.61 pF for the gate oxide, the measured value was 8.94 pF for a margin of error of 67%. Meanwhile, the field oxide measured at 1.5 pF, while it was expected to be 6.903 pF for a margin of error of 78%. The intermediate oxide had a theoretical value of 13.81 pF, while the measured value was 3.57 pF, which is also much lower than expected, having a 74% margin of error. These low values were likely due to poor timing and human error when it came to the oxide thicknesses. It is clear that the field oxide

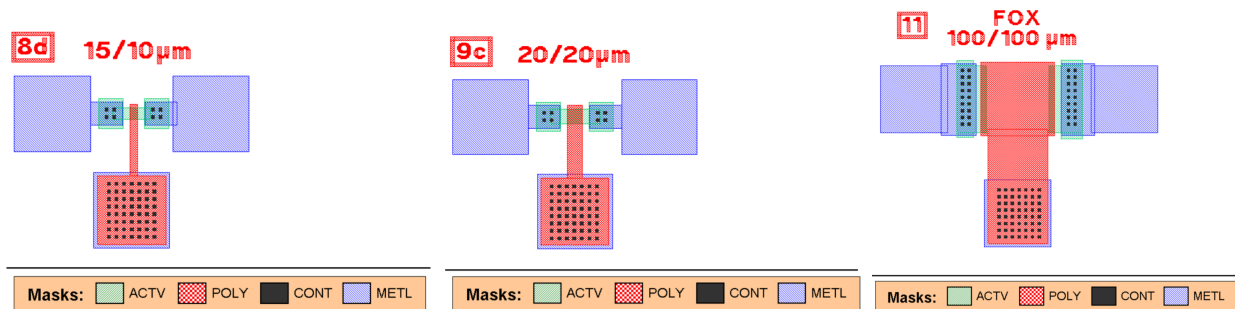
should have been much thicker, but because it is not, it is capable of affecting the performance of all other devices due to how thin the layer is.

VI. MOSFETs (8a, 8b, 8c, 8d, 9a, 9b, 9c, 10, 11)

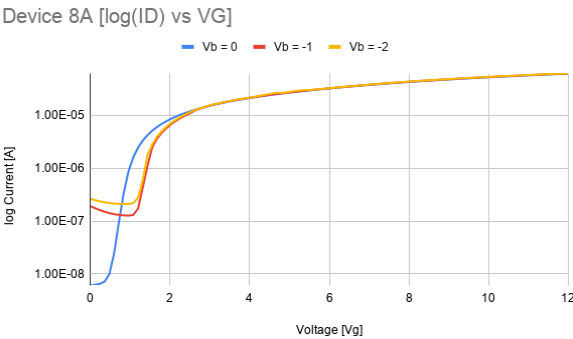
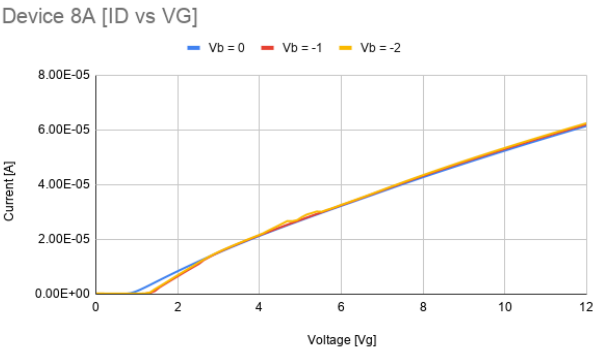
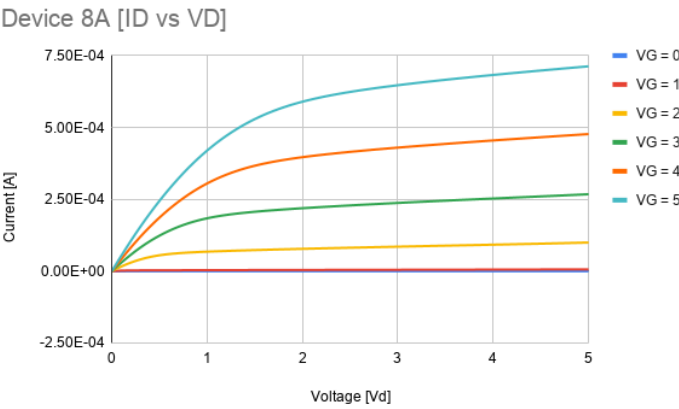
Overview:

A MOSFET is a semiconductor device that is in charge of manipulating electric signals and controlling the current between the gate and source. The probes would be placed onto the gate, source, and drain, with the remaining probe being on the chuck. The difference between 8a, 8b, 8c, and 8d is the length of the channels, because while 8a is $4\mu\text{m}$, 8d is $10\mu\text{m}$. This means that 8a has the capacity to increase current and switch speed because the resistance is lower. Meanwhile, 9a-9c differ by the widths, increasing by $5\mu\text{m}$, which directly makes 9c at $20/20\mu\text{m}$ have the highest current. Whilst 10 expresses a very wide and long MOSFET capable of high drive current design, whereas 11 is in charge of controlling and preventing leakage, which performance enhances differently from 10.

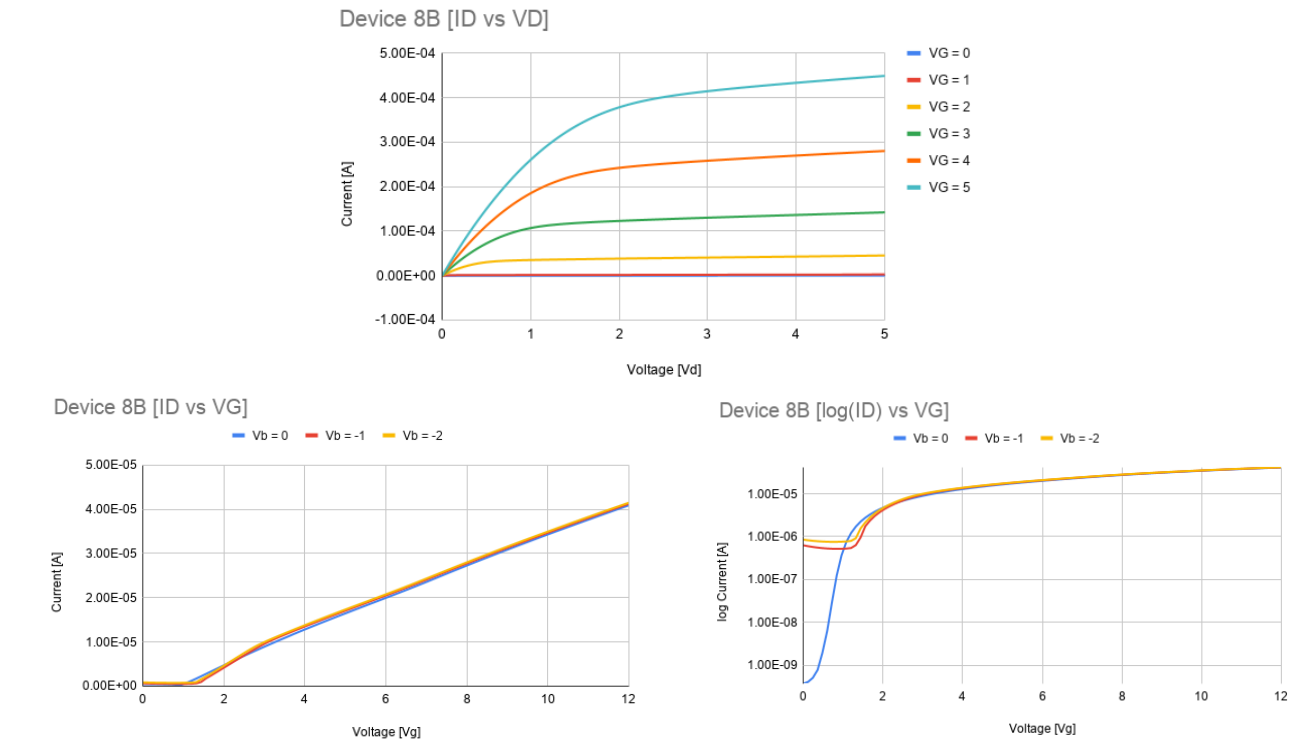
Layout:



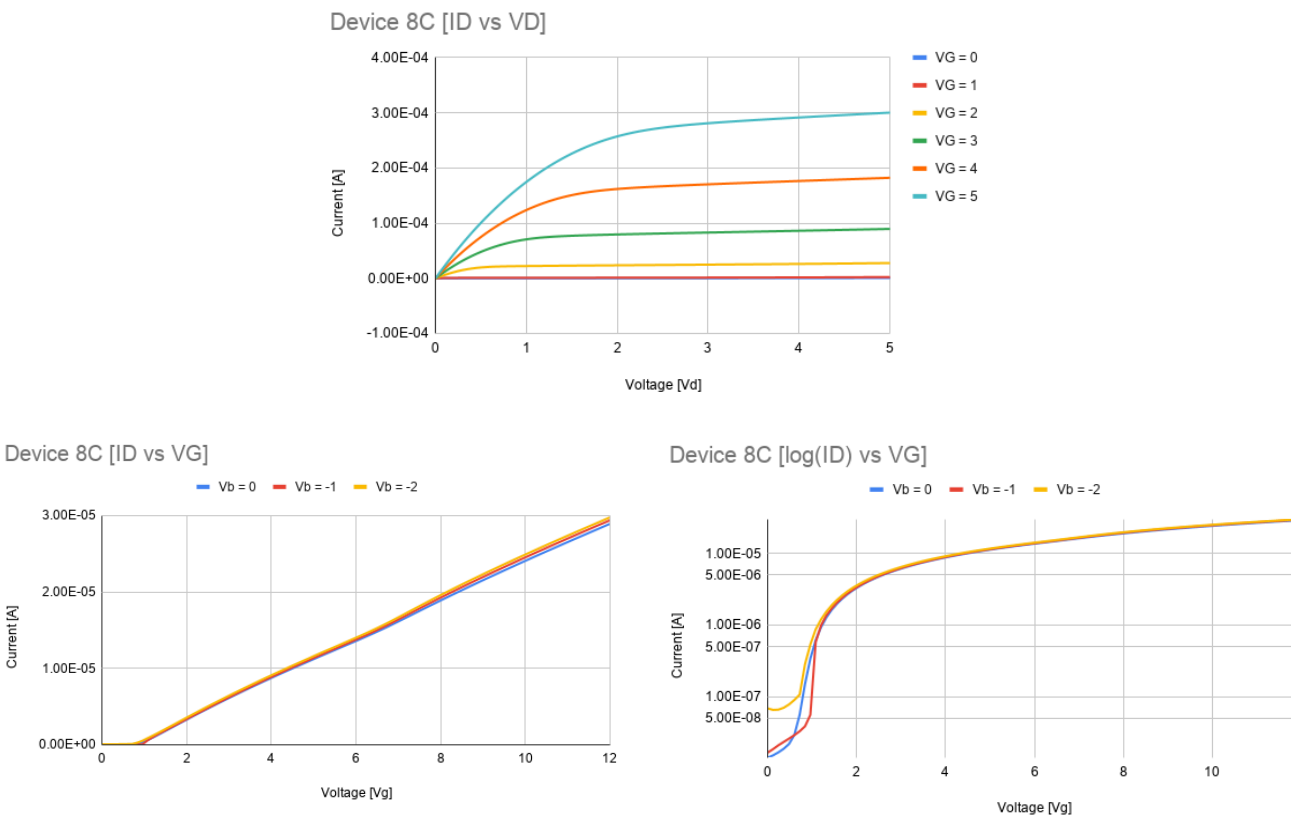
Measurements:
 8a:



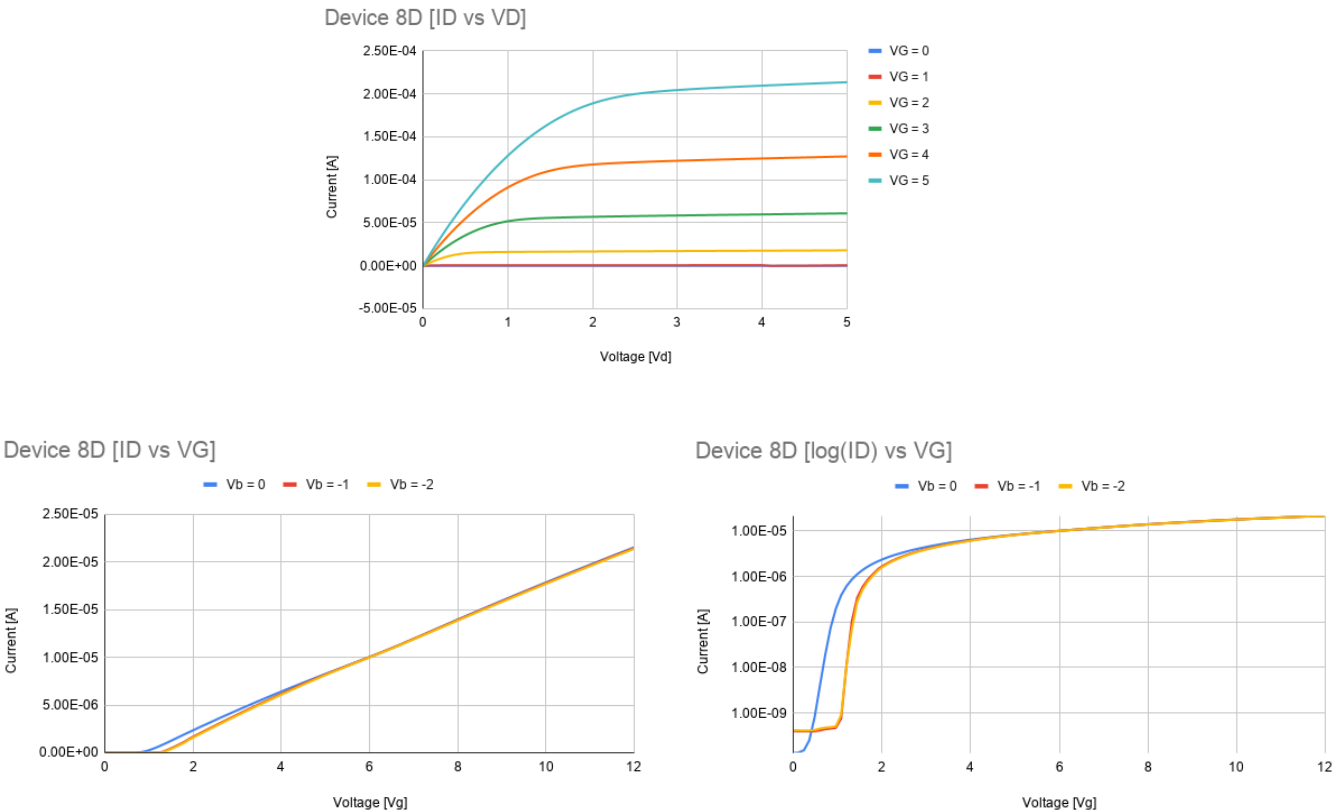
8b:



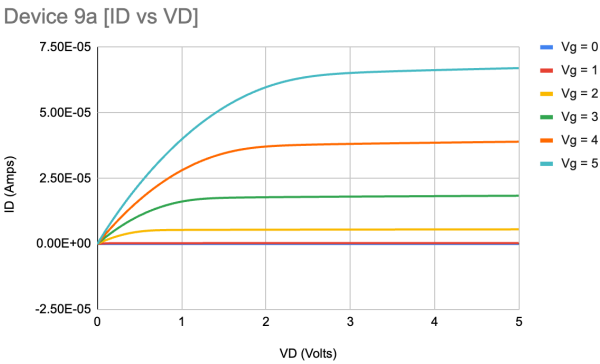
8c:

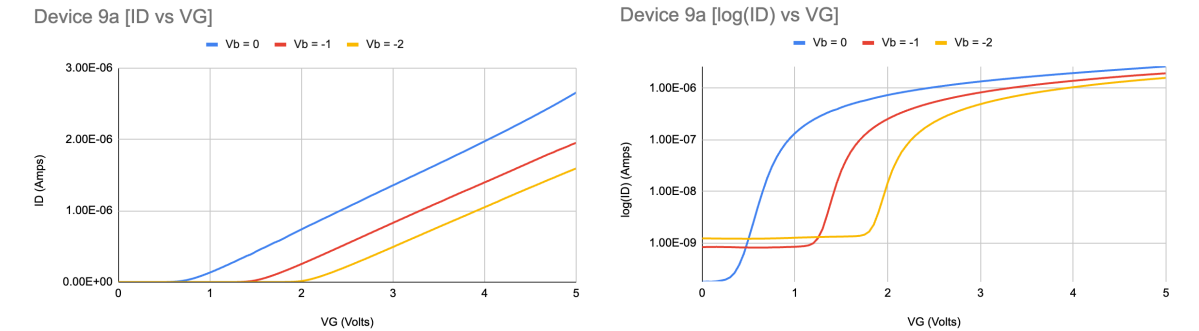


8d:

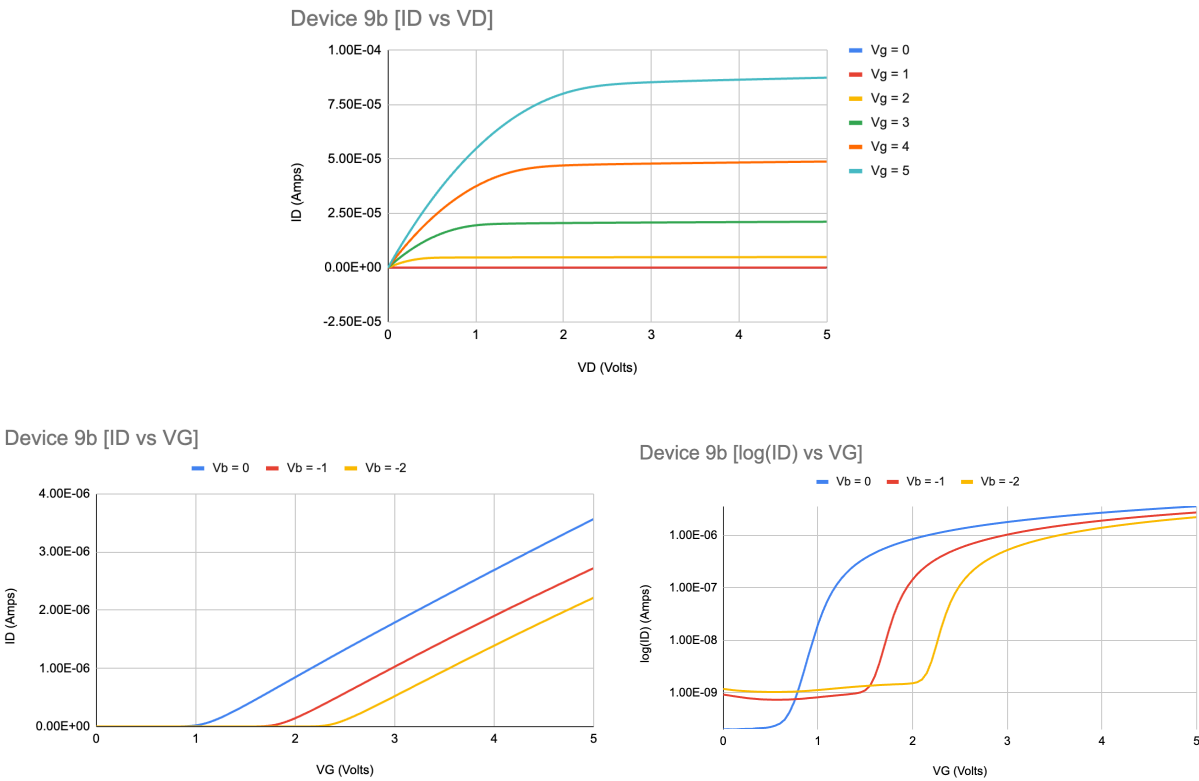


9a:

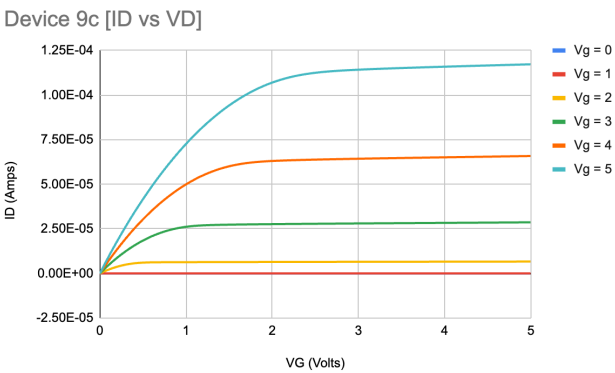




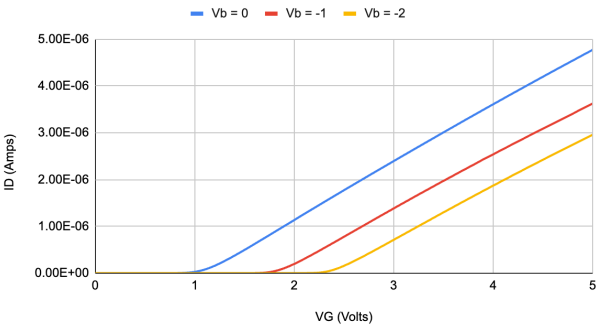
9b:



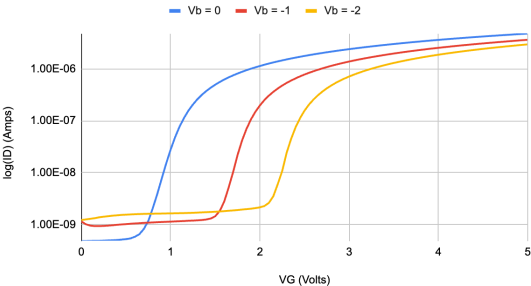
9c:



Device 9c [ID vs VG]

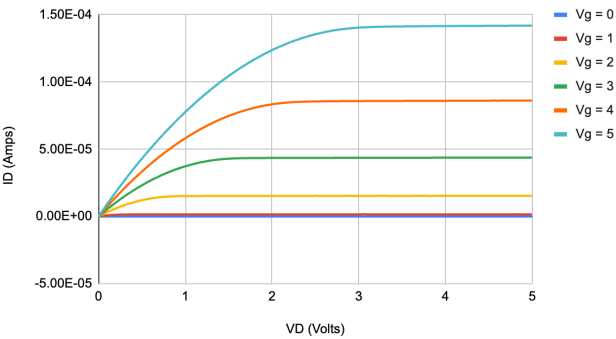


Device 9c [log(ID) vs VG]

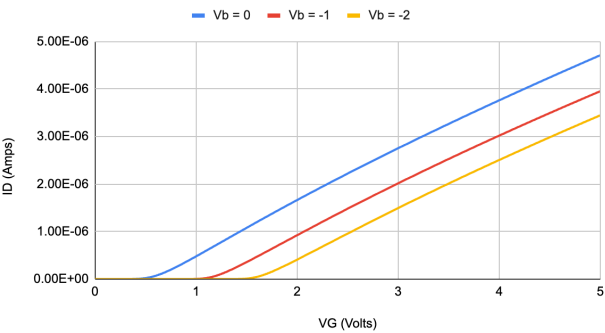


10:

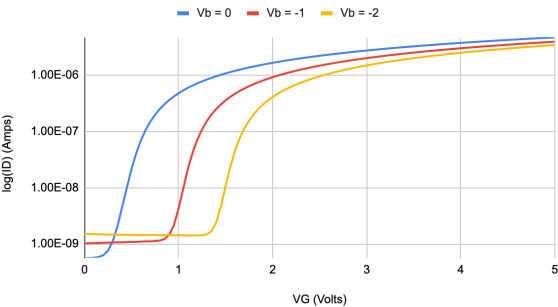
Device 10 [ID vs VD]



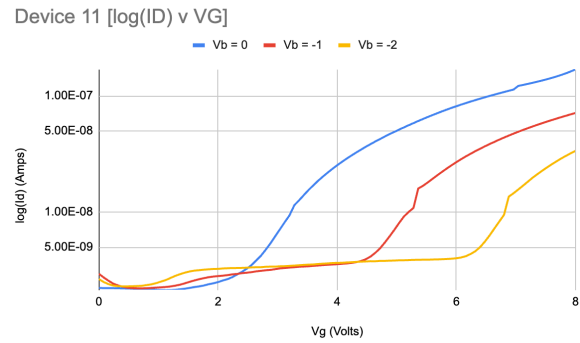
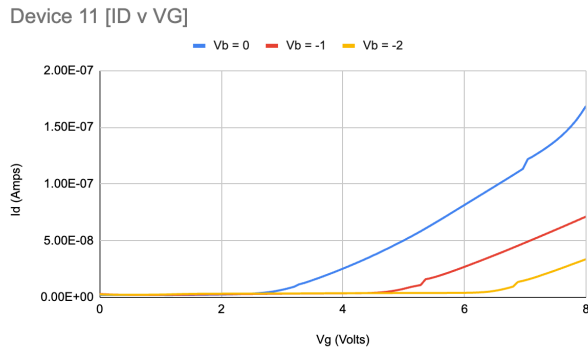
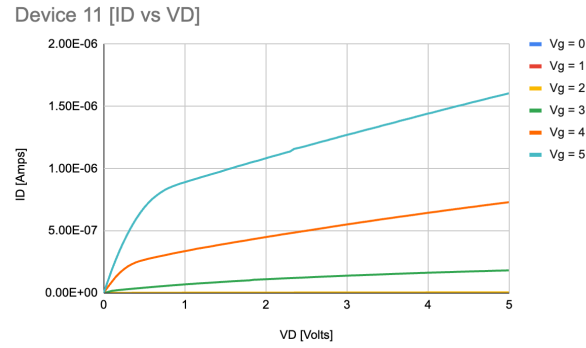
Device 10 [ID vs VG]



Device 10 [log(ID) vs VG]



11:



Device	V_T (V)	I_{off} (μA)	I_{on} (μA)
8a	0.588	6.07E-3	61.5
8b	0.422	3.74E-4	40.9
8c	0.514	1.42E-2	28.9
8d	0.504	1.32E-4	21.5
9a	0.52	1.76E-04	2.66
9b	0.776	2.01E-4	3.57
9c	0.771	4.85E-4	4.77
10	0.3267	5.59E-4	4.71
11	0.5019	2.23E-3	1.69E-1

V_T calculations explained in appendix

Calculations:

Device	8a	8b	8c	8d	9a	9b	9c	10	11
I_{on}/I_{off}	10132	109358	2035	160448	15114	17761	9835	8426	757.85

Yield (%) = ((Good devices)/(Total Devices))*100 = (9/9)*100 = 100%

Yield (%)	100%
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Discussion:

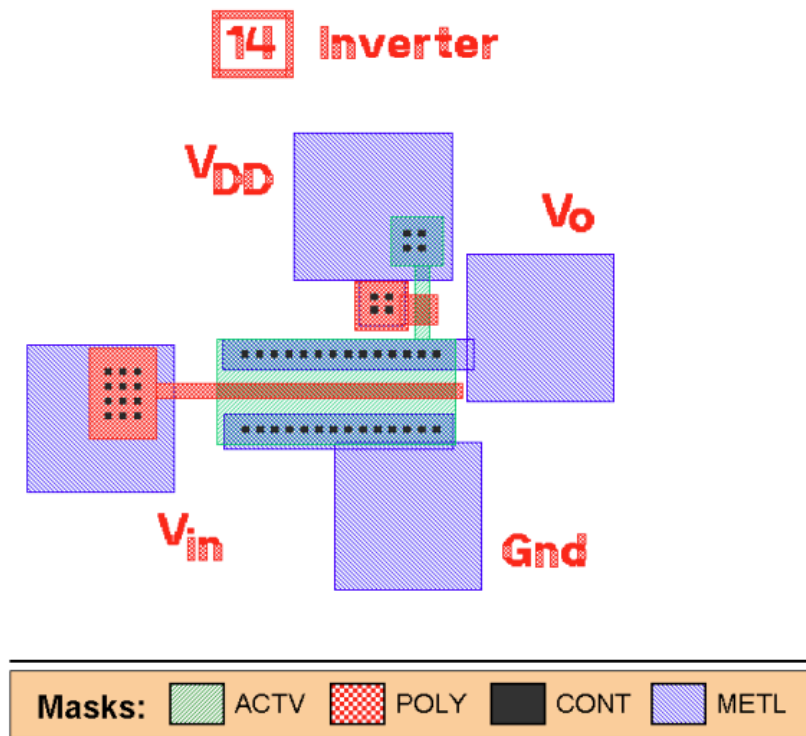
The general behavior of the MOSFETS aligned with device trends that the smaller area meant that the current switch happened earlier, and generally reported that the resistances were lower. Specifically, it was evident that we could get V_T calculation from the graphs that we got, which signified that the devices are functional. The yield represented that all the devices worked, and this occurred because the graphs that we got did have linear growth, which allowed for a slope to be attained. As we can see, MOSFET 11 does not follow the general order of magnitude and is much lower compared to other ratios, but does follow the expected behavior in trend and is functional. Overall, there was still a trendline that could be attained and a slope attributed to it.

VII. Inverter (14)

Overview:

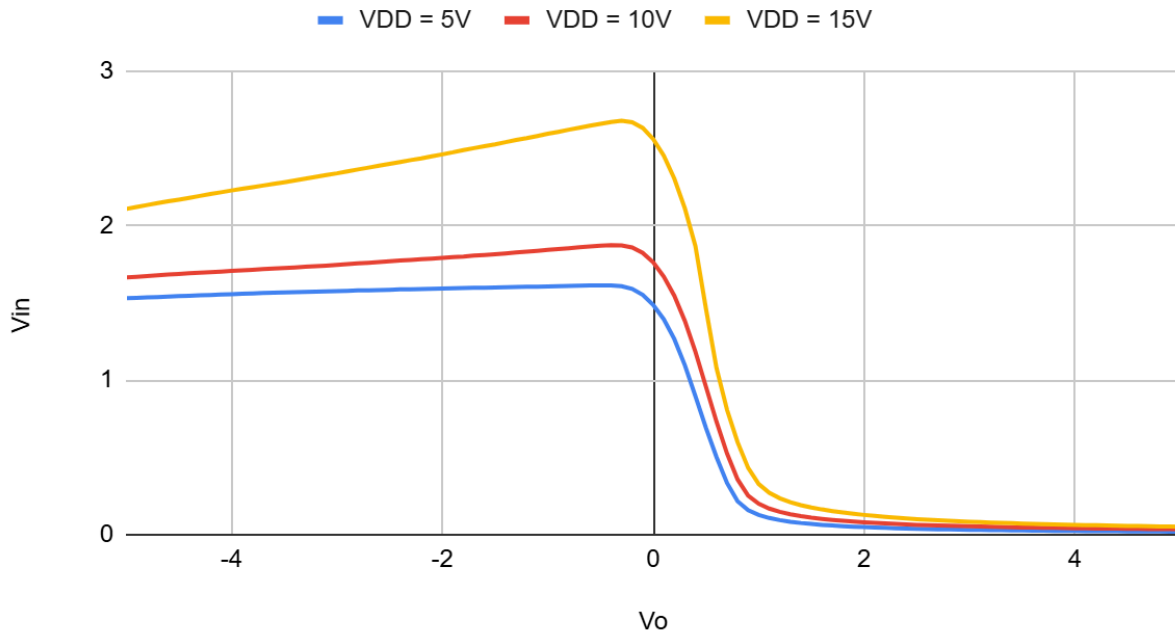
An inverter is designed to turn direct current into alternating current through the high-speed switching devices (8a-11) in order to achieve a source-to-drain current flow. This is measured by placing a probe on the drain, another on the ground, then the source, and lastly on the body/substrate. Then measure the voltage on all probes except the body/substrate where the current measurement is required.

Layout:



Measurements:

Vo vs. Vin varying VDD



Parameter	Value
V_o when $V_{in} = 0V$ (V)	1.935 V
V_o when $V_{in} =$ the above value of V_o (V)	-1.93 V

Discussion:

The inverter does not work. We tested approximately 10 different inverters and received similar poor results. It proved not to function properly, and the results did not adequately reflect the intended theoretical graph results. The digital logic would likely not work all things considered, but if it did, then it would be capable of inverting the results back. If I had the capacity to change the etching and diffusion process into a less contaminated space and had taken things much slower to ensure accuracy, that would be extremely helpful. Additionally, the aluminum section would have been done with more reliable equipment so that the resistors would be reliable and the thicknesses of the layers would be dependable.

Appendix

Measured:

$$2A: R = \frac{0.30495 - (-0.204968)}{0.002 - (-0.002)} = 102.48 \Omega$$

$$2B: R = \frac{0.444264 - (-0.445130)}{0.002 - (-0.002)} = 229.3 \Omega$$

Theoretical:

Isotropic etch so **overetch** = 0.5 μm per side = 1 μm total

$$2A) L = 200 + 1 = 201 \mu\text{m} \quad H = 1 \mu\text{m}$$

$$W = 10 + 1 = 11 \mu\text{m}$$

$$R_{2A} = R_s \frac{L}{W} = 3 \frac{201}{11} = 54.8 \Omega$$

$$2B) L = 200 - 1 = 199 \mu\text{m} \quad H = 0.45 \mu\text{m}$$

$$W = 10 - 1 = 9 \mu\text{m}$$

$$R_{2B} = R_s \frac{L}{W} = 3 \frac{199}{9} = 66.3 \Omega$$

Measured:

$$2C: R = \frac{1}{0.01000366} = 99.96 \Omega$$

$$2D: R = \frac{1}{0.00634354} = 157.64 \Omega$$

Theoretical: $R = R_s \frac{L}{W}$

$$R_{\text{metal}} = 0.3 \frac{150}{50} = 0.9 \Omega \quad R_{\text{active}} = 3 \frac{150}{50} = 9 \Omega \quad R_{\text{poly}} = 3 \frac{150}{50} = 9 \Omega$$

$$R_{2C, \text{no contacts}} = 7(R_{\text{metal}} + R_{\text{active}}) = 7(0.9 + 9) = 69.3 \Omega$$

$$R_{2d, \text{no contacts}} = 7(R_{\text{metal}} + R_{\text{poly}}) = 7(0.9 + 9) = 69.3 \Omega$$

$$R_{\text{single de contact}} = \frac{R_{2C, \text{measured}} - R_{2C, \text{no contacts}}}{14} = \frac{99.96 - 69.3}{14} = 2.19 \Omega$$

$$R_{\text{single 2d contact}} = \frac{R_{2d, \text{measured}} - R_{2d, \text{no contacts}}}{14} = \frac{157.64 - 69.3}{14} = 6.31 \Omega$$

Areas from layout:

$$\text{Device 3 (field oxide): } 200 \times 200 = 40000 \mu\text{m}^2$$

$$\text{Device 4 (gate oxide): } 200 \times 200 = 40000 \mu\text{m}^2$$

$$\text{Device 5 (intermediate oxide): } 200 \times 200 = 40000 \mu\text{m}^2$$

$$A = 4 \times 10^4 \times 10^{-12} = 4 \times 10^{-8} \text{ m}^2$$

$$\text{Field oxide: } 200 \text{ nm} = 2 \times 10^{-7} \text{ m}$$

$$\text{Gate oxide: } 50 \text{ nm} = 5 \times 10^{-8} \text{ m}$$

$$\text{Intermediate oxide: } 100 \text{ nm} = 1 \times 10^{-7} \text{ m}$$

$$C = \frac{\epsilon_0 \epsilon_r A}{t}$$

$$C = \frac{(8.85 \times 10^{-12})(3.9)(4 \times 10^{-8})}{2 \times 10^{-7}} = 6.903 \times 10^{-12} = 6.903 \text{ pF}$$

$$C = \frac{(8.85 \times 10^{-12})(3.9)(4 \times 10^{-8})}{5 \times 10^{-8}} = 2.761 \times 10^{-11} = 27.61 \text{ pF}$$

$$C = \frac{(8.85 \times 10^{-12})(3.9)(4 \times 10^{-8})}{1 \times 10^{-7}} = 1.381 \times 10^{-11} = 13.81 \text{ pF}$$

Measured:

Forward Threshold Voltage: $\sim 0.6 \text{ V}$

Breakdown Voltage: $\sim 37.6 \text{ V}$

Theoretical: (Assuming room temp)

$$N_A = \frac{1}{q \mu_p P} = \frac{1}{(1.6 \times 10^{-19})(480)(1)} = 1.3 \times 10^{16} \text{ cm}^{-3}$$

$$N_D = \frac{1}{q \mu_n P} = \frac{1}{(1.6 \times 10^{-19})(1350)(10^{-3})} = 4.6 \times 10^{18} \text{ cm}^{-3}$$

$$V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right) = 0.0259 \ln \left(\frac{(1.3 \times 10^{16})(4.6 \times 10^{18})}{(10^{10})^2} \right) = 0.88 \text{ V}$$

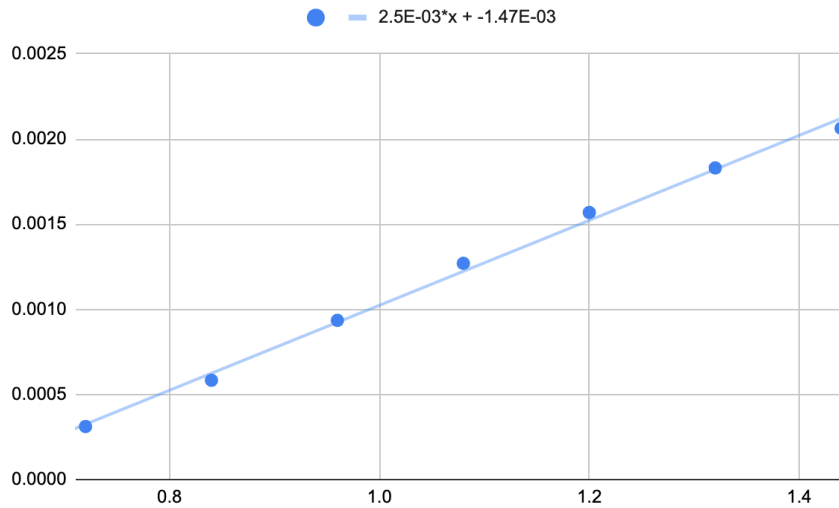
Graphs for V_t calculations (using the linear region of $\sqrt{I_D}$ vs V_G)

In order to extract the threshold voltage, we used the relationship that $I_D \propto (V_G - V_T)^2$ in the linear region. Thus, if we plot a graph of $\sqrt{I_D}$ vs V_G we would get a relationship of

$\sqrt{I_D} = m(V_G - V_T) = mV_G - mV_T$. Thus, we could plot the graph and constrain our x-axis such that we are in the linear region (which was achieved through visual inspection to estimate where the linear region began), and fit a trendline using Google Sheets, and use the equation for the trendline to extract V_T . The intercept of the trendline (b) would be equal to mV_T so V_T would

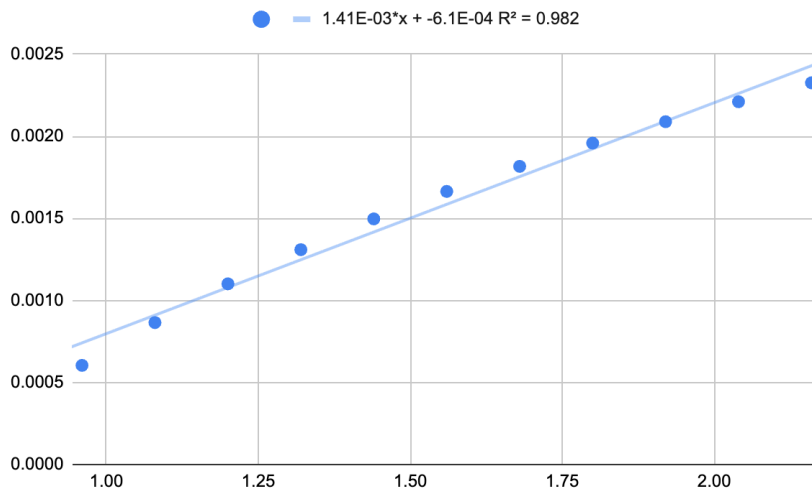
be $-b/m$ (the intercept over the slope). We ignored the subthreshold region and the region where we have mobility degradation.

8a)



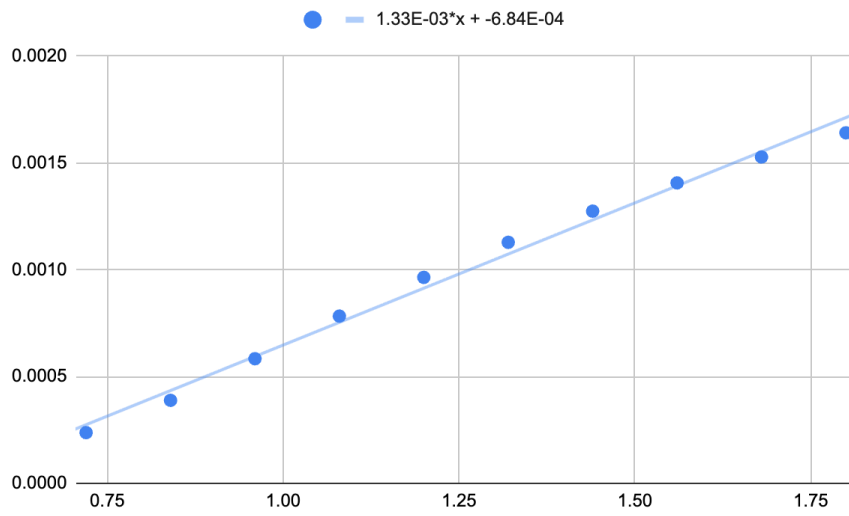
$$V_T = 1.47 * 10^{-3} / 2.5 * 10^{-3} = 0.588$$

8b)



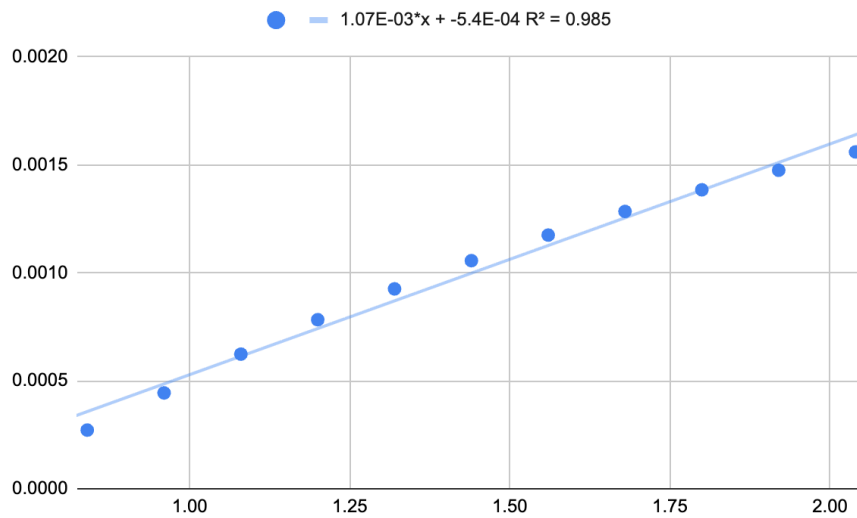
$$V_T = 6.1 * 10^{-4} / 1.41 * 10^{-3} = 0.422$$

8c)



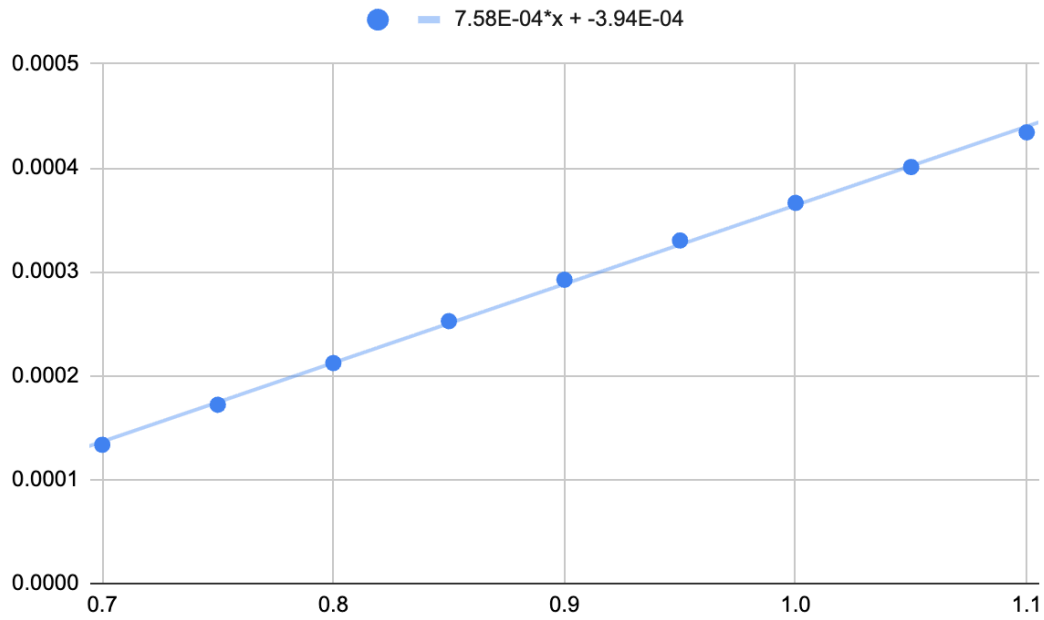
$$V_T = 6.84 * 10^{-4} / 1.33 * 10^{-3} = 0.514$$

8d)



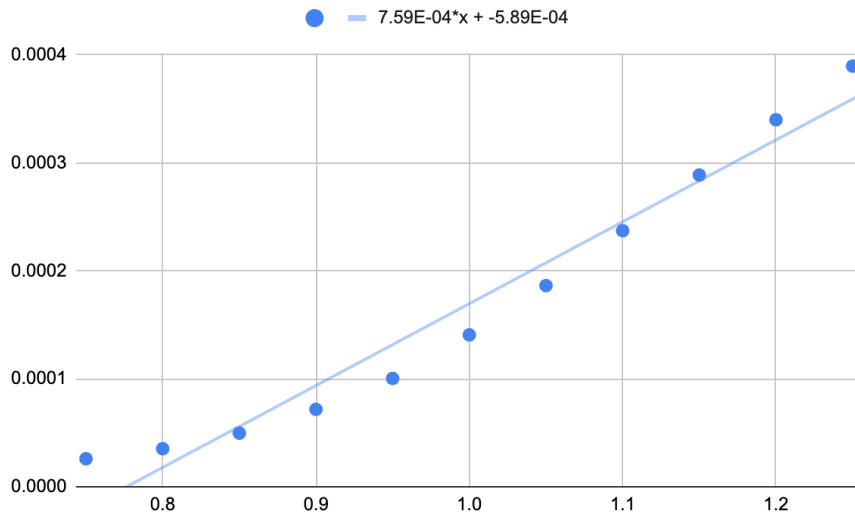
$$V_T = 5.4 * 10^{-4} / 1.07 * 10^{-3} = 0.504$$

9a)



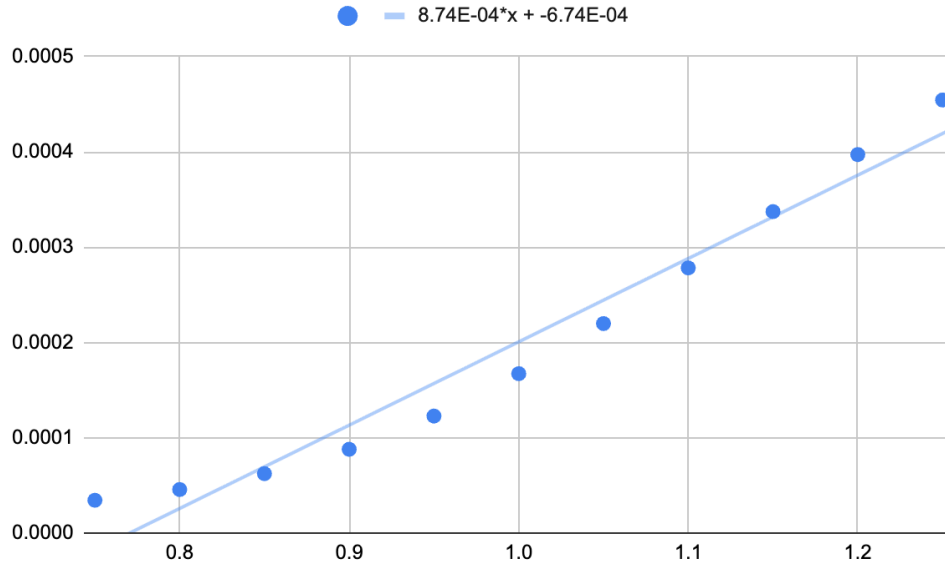
$$V_T = -b/m = 3.94 * 10^{-4} / 7.58 * 10^{-4} = 0.52$$

9b)



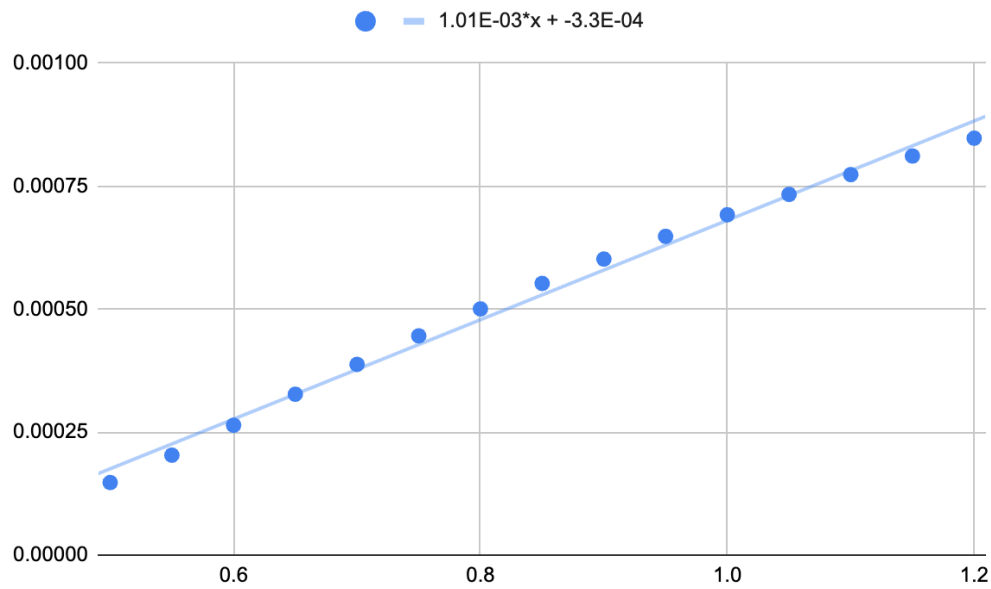
$$V_T = b/m = 5.89 * 10^{-4} / 7.59 * 10^{-4} = 0.776$$

9c)



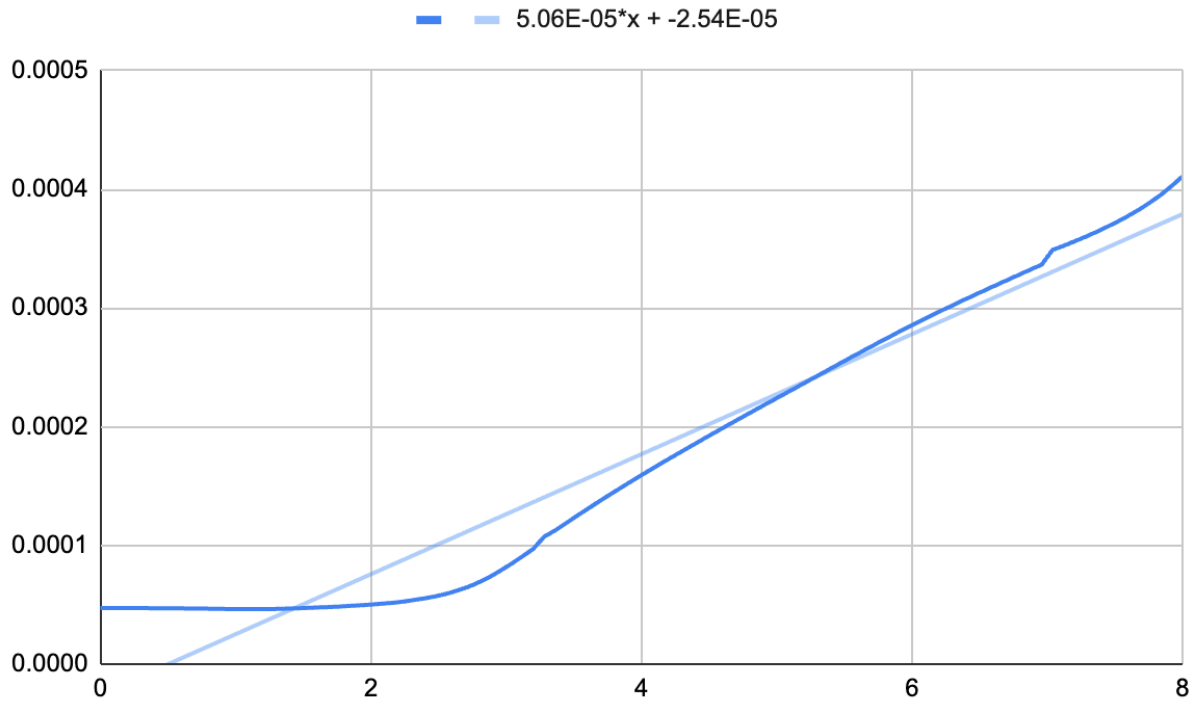
$$V_T = b/m = 6.74 * 10^{-4} / 8.74 * 10^{-4} = 0.771$$

10)



$$V_T = b/m = 3.3 * 10^{-4} / 1.01 * 10^{-3} = 0.3267$$

11)



$$V_T = b/m = 2.54 * 10^{-5} / 5.06 * 10^{-5} = 0.50197628458$$

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